Random Effects

PUBH 526

Design and Analysis of Randomized Trials in Public Health

Nilupa S. Gunaratna, MS, PhD

Fixed vs. Random Effects

- Fixed factor: Want to draw inference on the specific levels of that factor in your experiment/study
- Random factor: Want to draw inference on the population of factor levels
- Example: community tutoring program, analyze student performance data from 4 tutors
  - Are you interested in those 4 tutors?
  - Or are you interested in a larger population of tutors?
- In public health we are usually interested in larger populations
  - Participants are random effects
  - Interventions are fixed effects
  - Communities can be fixed or random depending on your objectives

Fixed vs. Random Effects

- This is a conceptual difference, and it affects your statistical model (and therefore also affects your analysis and interpretation)
- Rather than individual treatment effects (τ_i), we want to assess σ_τ^2
- Inference is broader with random effects – interested in a population of effects, not just the effects in your study
- You want to randomly select the effects in your study to represent that larger population (example: tutors; beware selection bias)

Random Effects Model (CRD)

- Same model as the fixed case with additional assumptions

$$y_{ij} = \mu + \tau_i + \epsilon_{ij} \quad \begin{cases} i = 1, 2 \ldots a \\ j = 1, 2, \ldots n_i \end{cases}$$

μ - grand mean

τ_i - ith treatment effect

$\epsilon_{ij} \sim \mathrm{N}(0, \sigma^2)$, independent

- Additionally assume

$\tau_i \sim \mathrm{N}(0, \sigma_\tau^2)$

$\{\tau_i\}$ and $\{\epsilon_{ij}\}$ independent

Students with the same tutor have correlated scores

- The results in $\mathrm{V}(y_{ij}) = \sigma_\tau^2 + \sigma^2$, $\mathrm{Cov}(y_{ij}, y_{ik}) = \sigma_\tau^2$

Inference

- The primary hypotheses are:

$$H_0 \ : \ \sigma_\tau^2 = 0$$
$$H_1 \ : \ \sigma_\tau^2 > 0$$

- Partitioning of Total Sum of Squares identical but expected mean squares change

  $\mathrm{E}(\mathrm{MS}_\mathrm{E}) = \sigma^2$

  $\mathrm{E}(\mathrm{MS}_\mathrm{Treatment}) = \sigma^2 + n\sigma_\tau^2$

- However, under H_0, the statistic $F_0 \sim F_{\alpha,a-1,N-a}$
- Thus, same test as before but different scope of inference
- Conclusions directed at the population of levels

Model Estimates (ANOVA Method)

- Usually interested in estimating variances
- Using mean squares from ANOVA table:

$$\hat{\sigma}^2 = \mathrm{MS_E}$$

$$\hat{\sigma}_\tau^2 = (\mathrm{MS_{Treatment}} - \mathrm{MS_E})/\mathrm{n}$$

If unbalanced, replace n with

$$n_0 = ((\sum n_i)^2 - \sum n_i^2)/((a-1)\sum n_i)$$

- However, using this approach, the estimate of σ_τ^2 can be negative → **maximum likelihood estimation**

“random villages.sas”

- Pay attention to variances and standard errors
- Note degrees of freedom for intercept
- Analogous to subsampling